Astro Night Planner 1.4.2

Practical manual for astrophotography planning with weather, Moon, object selection, equipment, horizon, framing and local surveys

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Contents

Overview

Astro Night Planner combines visibility, twilight, Moon, weather, location, horizon, equipment and Aladin framing in one planning interface. Profiles, equipment, locations and settings are stored locally in the browser. Create an external backup after important changes.

Version 1.4.2 adds integrated survey downloads, operating-system folder dialogs and a clear bulk switch for local sources. The Windows solution with automatically opened browser interface, tray and autostart and the native Linux/macOS edition remain available.

Important when updating to 1.4.2: Fully exit the previous ANP Local Survey Server from its tray menu and replace `ANP-Local-Survey-Server.exe` with the file from the **ANP Local Survey Server 1.1** package. The executable name remains unchanged. Open the complete version history.

Planning night, date buttons and Moon

Date buttons summarize the selected night. Weather and Moon values refer to the selected planning window, not automatically to the entire night.

Indicator	Meaning
↑	Moonrise, if it occurs in the relevant night window.
▲	Maximum Moon altitude **within the selected planning window**.
↓	Moonset. If it is outside the planning window, the planning data says so explicitly.

The planning data may also show the Moon culmination of the night. If it is before or after the planning window, the app adds a corresponding note.

Equipment and Aladin framing

Telescope, reducer/corrector/Barlow and camera determine effective focal length, field of view, pixel scale and the camera frame. In the Aladin view, combinations can be tried temporarily. If a combination is not saved as a setup, it still remains active and can be saved as a new setup.

Example: 700 mm focal length with a 0.80x reducer gives 560 mm effective focal length. With the same camera the field of view becomes larger.

Local surveys and Local Survey Server

Local surveys are local copies of HiPS sky maps. They are served through a local HTTP server, not through file paths. The app and Aladin expect URLs such as `http://127.0.0.1:8765/nsns/ohs8/properties`.

Common principle

All variants serve a shared survey root folder. The app configuration is the same on every operating system:

```
Local survey base URL: http://127.0.0.1:8765/
Relative path:         nsns/ohs8/
```

The root folder may be `D:\AstroSurveys`, `/home/user/AstroSurveys` or `/Users/user/AstroSurveys`.

Windows variant

The executable intentionally has no version number in its filename: `ANP-Local-Survey-Server.exe`. The version is shown in the ZIP package and interface, keeping shortcuts and autostart entries stable.

On Windows, the **ANP Local Survey Server 1.1** package is recommended. Double-clicking the EXE automatically opens the browser configuration interface; tray menu, autostart and background mode remain available.

Action	Function
Open interface	Opens the local configuration page.
Start server	Starts the survey file server at the configured base URL.
Stop server	Stops only the survey file server. The helper remains active.
Exit	Stops everything completely and releases the EXE for replacement or deletion.

Close browser window	Does not exit the program. Server and tray keep running; double-click the tray icon to reopen the interface.

Folder selection and migration of the previous configuration

On Windows, server 1.1 uses a native Windows folder picker. PowerShell is not required. On first start the program also searches for the previous `ANP-Local-Survey-Server-config.json` in the program folder, current folder, user folder and AppData. If found, survey root, port, language and autostart settings are migrated. The old file remains as a backup.

Downloading other HiPS surveys

The download manager includes DSS2, PanSTARRS, VTSS, Finkbeiner, WISE, 2MASS, IRIS and all six NSNS surveys. In addition, select **Other / custom HiPS survey** for another source. Then enter a complete HiPS source URL and a relative target path inside the survey root, for example `custom/my-survey/`.

> **Functional limit:** A complete recursive download is possible only when the source server provides a directory listing and the download is technically and legally permitted. Very large surveys may require substantial storage and download time.

Saving without a popup

After a successful save, the Save button briefly changes colour and text. A separate success popup is no longer shown. Errors are still displayed directly with the configuration.

Linux/macOS variant

For Linux and macOS, use **ANP Local Survey Server Linux/macOS 1.1**. It is a native server solution with browser interface. Wine and Python are not required.

Included binaries:

System	File
Linux Intel/AMD 64-bit	

	`bin/anp-local-survey-server-linux-amd64`
Linux ARM64	`bin/anp-local-survey-server-linux-arm64`
macOS Intel	`bin/anp-local-survey-server-darwin-amd64`
macOS Apple Silicon	`bin/anp-local-survey-server-darwin-arm64`

Start on Linux

```
chmod +x bin/anp-local-survey-server-linux-amd64
./bin/anp-local-survey-server-linux-amd64 --root "$HOME/
AstroSurveys" --port 8765 --open
```

Start on macOS

```
chmod +x bin/anp-local-survey-server-darwin-arm64
./bin/anp-local-survey-server-darwin-arm64 --root "$HOME/
AstroSurveys" --port 8765 --open
```

On first launch, macOS may require a security approval because the package is not from the App Store. The configuration interface normally opens at `http://127.0.0.1:8776/`.

Autostart

On Linux, use a `systemd --user` service. The package contains `templates/anp-local-survey-server.service`. On macOS, use a LaunchAgent. The package contains `templates/de.deepskyastrophoto.anp-local-survey-server.plist`.

Python fallback

The Python package remains available as a transparent fallback or expert solution. It is useful if a user prefers a small script or if the native variant does not start on a special platform.

Folder structure

Folder below the root folder	Relative path in Planner	Meaning
`nsns/ohs8`	`nsns/ohs8/`	

		NSNS OIII/H-alpha/SII composite
`nsns/halpha8`	`nsns/halpha8/`	NSNS H-alpha
`nsns/oiii8`	`nsns/oiii8/`	NSNS OIII
`nsns/sii8`	`nsns/sii8/`	NSNS SII
`nsns/hbr8`	`nsns/hbr8/`	NSNS H-beta/red, if available
`nsns/rgb8`	`nsns/rgb8/`	NSNS RGB, if available

HiPS content and tile format

A HiPS folder must contain a `properties` file. For display, `Norder...`, `Dir...` and `Npix...` are also required. The `properties` file contains `hips_tile_format`. If it says `png`, PNG tiles must be used; if it says `jpeg`, JPEG tiles must be used.

Folder selection

In the Local Survey Server browser interface, **Choose folder** opens an operating-system dialog for the survey root. The download target can also be selected and must remain inside the root. A normal file dialog is not suitable for server-relative paths in Astro Night Planner. The button next to a relative path therefore queries the running server and lists detected HiPS folders containing a `properties` file.

Download survey data

Download and manage survey data opens the running Local Survey Server download manager. Known NSNS surveys can be selected and recursively copied from the online directory into the local root. Complete existing files are skipped and partial files can be resumed.

Function	Effect
Start download	Starts recursive download into the relative target path.
Pause/Resume	Pauses or resumes the active download.
Cancel	

	Ends the run; existing partial and target files remain for later continuation.
Validate existing data	Checks `properties`, tile format and existing file count.

The source server must provide a readable directory listing. Large surveys require sufficient storage and a stable internet connection.

Global and individual preference

Prefer local sources for all surveys is a bulk switch. It sets or clears **Prefer local source** in every survey row. Individual surveys can then be changed separately. Local survey usage itself must also be enabled.

Fallback and diagnostics

The app first checks the local source when local use and local preference are enabled. If reachable, the local URL is passed to Aladin. If it is unavailable and online fallback is enabled, the online source is used. If fallback is disabled, an error message is shown.

> **Important:** The **Open Local Survey Server** button cannot start a local program. It only opens the configuration interface if the matching server is already running.

Weather, cloud model and external sources

Weather, cloud model, Meteoblue, Windy, Ventusky and other external reference sources support the decision whether a night is suitable for astrophotography. External services require an internet connection.

Backup, PWA and troubleshooting

The app uses local browser data. Create regular full backups. Local survey requests to `127.0.0.1`, `localhost` and private LAN addresses are not intercepted by the PWA cache but passed directly to the local server.

- If local surveys do not work, first check `http://127.0.0.1:8765/nsns/ohs8/properties` in the browser.
- Before replacing the Windows server package, use **Exit** in the tray menu.
- If Linux/macOS should start in the background, configure systemd user service or LaunchAgent.

Glossary

Term	Explanation
HiPS	Hierarchical Progressive Survey, a tiled sky-map format.
properties	Control file of a HiPS survey with tile format and resolution.
Norder	Folder structure of HiPS tiles by resolution order.
Fallback	Switching to the online source if the local source is not reachable and fallback is enabled.